

Department of Computer Engineering

University of Peradeniya

Assignment: Basics of Software Testing

CO328 : Software Engineering

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Part 1

A. Basics of Software Testing

1. What is the purpose of software testing?
2. Compare and contrast manual testing vs automated testing.
3. What are functional and non-functional testing? Give examples of each.

B. Testing Levels

4. Explain the four main levels of testing with an example for each.
5. What's the difference between system testing and acceptance testing?

C. Unit Testing

6. What is unit testing and why is it important?
7. Write a simple Java method `multiply(int a, int b)` and a JUnit test for it.
8. What are the benefits and limitations of unit testing?

D. JUnit

9. What do the following JUnit annotations do: `@Test`, `@Before`, `@After`, `@Ignore`?
10. Convert this method to a JUnit test:

```
public int divide(int a, int b) {  
    return a / b;  
}
```

E. Selenium

11. What is Selenium used for?
12. Write a simple Selenium script in Java that opens Google and searches for "JUnit tutorial".
13. What are locators in Selenium? List at least three types with examples.
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Part 2

Task 1: Testing Types Table

Fill in the table below:

Test Type	Functional/Non-Functional	Description	Tool Used
Unit Testing	?	Tests individual methods	?
Performance Testing	?	Measures response time	?
System Testing	?	Tests entire application	?
Security Testing	?	Validates data safety	?
UI Testing	?	Tests user interface behavior	?

Task 2: Testing Levels Table

Complete this with examples:

Level	Example Tool	What It Tests
Unit Testing		
Integration Testing		
System Testing		
Acceptance Testing		

Part 3

Task A: JUnit Practice

1. Create a class `Calculator` with methods `add`, `subtract`, `multiply`, `divide`.
2. Write a JUnit test class `CalculatorTest` and test all four methods.
3. Add a test that checks for division by zero using `assertThrows`.

Task B: Selenium Practice

1. Write a Selenium script that:
 - Opens a browser.
 - Goes to "<https://example.com>".
 - Verifies that the page title is "Example Domain".
 2. Add code to take a screenshot and save it locally.
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Part 4

1. What is regression testing, and when should it be done?
2. Explain white-box testing with an example.

3. Describe the defect lifecycle with a real-world analogy.
4. What is CI/CD? How is testing integrated into the pipeline?
5. Why is it important to write both positive and negative test cases?